

Day3_Assessment

CASE STUDY 1 — Online Shopping Order System:

```
>_MONGOSH
> db.createCollection("orders")
< { ok: 1 }
> db.orders.insertMany([
  {customer_id: "C101",
    name: "Asha",
    items: [{ product: "Mobile", price: 15000 },
      { product: "Earphones", price: 2000 }
    ]
  },
  {
    customer_id: "C102",
    name: "Kiran",
    items: [
      { product: "Mobile", price: 15000 },
      { product: "Laptop", price: 50000 }
    ]
  },
  {
    customer_id: "C103",
    name: "Neha",
    items: [
      { product: "Earphones", price: 2000 },
      { product: "Laptop", price: 50000 }
    ]
  }
])
< {
  acknowledged: true,
  insertedIds: {
```

>_MONGOSH

```
> db.orders.aggregate([
  { $unwind: "$items" },

  {
    $match: {
      "items.product": "Mobile",
      "items.price": { $gte: 15000 }
    }
  },

  {
    $group: {
      _id: "$items.product",
      totalCustomers: { $sum: 1 },
      totalRevenue: { $sum: "$items.price" },
      averagePrice: { $avg: "$items.price" },
      minimumPrice: { $min: "$items.price" },
      maximumPrice: { $max: "$items.price" },
      customerNames: { $push: "$name" },
      firstCustomer: { $first: "$name" }
    }
  },

  {
    $out: "mobile_report"
  }
])
```

<

```
> db.mobile_report.find()
< {
  _id: 'Mobile',
  totalCustomers: 2,
  totalRevenue: 30000,
  averagePrice: 15000,
  minimumPrice: 15000,
  maximumPrice: 15000,
  customerNames: [
    'Asha',
    'Kiran'
  ],
  firstCustomer: 'Asha'
}
```

>_MONGOSH

```
> db.orders.aggregate([
  { $unwind: "$items" },
  { $match: { "items.price": { $gte: 15000 },
    "items.product": { $in: ["Mobile", "Laptop"] }
  }},
  { $group: { _id: "$items.product",
    totalOrders: { $sum: 1 },
    totalRevenue: { $sum: "$items.price" },
    averagePrice: { $avg: "$items.price" },
    minimumPrice: { $min: "$items.price" },
    maximumPrice: { $max: "$items.price" },
    customerList: { $push: "$name" }
  }},
  { $out: "high_value_products_report" }
])
<
> db.high_value_products_report.find()
< {
  _id: 'Laptop',
  totalOrders: 2,
  totalRevenue: 100000,
  averagePrice: 50000,
  minimumPrice: 50000,
  maximumPrice: 50000,
  customerList: [
    'Kiran',
    'Neha'
  ]
}
```

>_MONGOSH

```
> db.orders.aggregate([
  { $unwind: "$items" },
  {$group: {
    _id: "$name",
    totalSpent: { $sum: "$items.price" },
    averagePrice: { $avg: "$items.price" },
    minimumPrice: { $min: "$items.price" },
    maximumPrice: { $max: "$items.price" },
    productList: { $push: "$items.product" },
    totalProducts: { $sum: 1 } }},
  {$match: {
    totalProducts: { $gt: 1 },
    totalSpent: { $gte: 30000 } }},
  {$out: "customer_spending_report"}
])
<
> db.customer_spending_report.find()
< {
  _id: 'Kiran',
  totalSpent: 65000,
  averagePrice: 32500,
  minimumPrice: 15000,
  maximumPrice: 50000,
  productList: [
    'Mobile',
    'Laptop'
  ],
  totalProducts: 2
}
```

CASE STUDY 2 — Movie Ticket Booking System

```
>_MONGOSH

> db.createCollection("customers")
< { ok: 1 }

> db.customers.insertMany([
  { _id: 1,
    name: "Arun",
    profile: { age: 25, city: "Chennai" } },
  { _id: 2,
    name: "Priya",
    profile: { age: 22, city: "Bangalore" } },
  { _id: 3,
    name: "Rahul",
    profile: { age: 30, city: "Hyderabad" } },
  { _id: 4,
    name: "Neha",
    profile: { age: 28, city: "Mumbai" } },
  { _id: 5,
    name: "Kiran",
    profile: { age: 35, city: "Pune" } }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': 1,
    '1': 2,
    '2': 3,
    '3': 4,
    '4': 5
  }
}
```

>_MONGOSH

```
> db.movies.insertMany([
  { _id: "M1", title: "Inception", genre: "Sci-Fi", price: 300 },
  { _id: "M2", title: "Avatar", genre: "Fantasy", price: 350 },
  { _id: "M3", title: "Interstellar", genre: "Sci-Fi", price: 400 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': 'M1',
    '1': 'M2',
    '2': 'M3'
  }
}

> db.bookings.insertMany([
  { _id: 301, customer_id: 1, movie_id: "M1", show_date: "2026-02-01" },
  { _id: 302, customer_id: 1, movie_id: "M2", show_date: "2026-02-03" },
  { _id: 303, customer_id: 2, movie_id: "M1", show_date: "2026-02-05" },
  { _id: 304, customer_id: 3, movie_id: "M3", show_date: "2026-02-06" },
  { _id: 305, customer_id: 3, movie_id: "M1", show_date: "2026-02-08" },
  { _id: 306, customer_id: 4, movie_id: "M2", show_date: "2026-02-10" },
  { _id: 307, customer_id: 5, movie_id: "M3", show_date: "2026-02-12" },
  { _id: 308, customer_id: 5, movie_id: "M2", show_date: "2026-02-14" }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': 301,
    '1': 302,
    '2': 303,
```

>_MONGOSH

```
> db.bookings.aggregate([
  {$group: {
    _id: "$movie_id",
    totalBookings: { $sum: 1 }}}
])
< {
  _id: 'M3',
  totalBookings: 2
}
{
  _id: 'M1',
  totalBookings: 3
}
{
  _id: 'M2',
  totalBookings: 3
}
day2> |
```


>_MONGOSH

```
> db.bookings.aggregate([
  {$lookup: {
    from: "movies",
    localField: "movie_id",
    foreignField: "_id",
    as: "movieDetails"}}},
  { $unwind: "$movieDetails" },
  {$group: {_id: "$movieDetails.title",
    totalTicketsSold: { $sum: 1 },
    totalRevenue: { $sum: "$movieDetails.price" },
    averagePrice: { $avg: "$movieDetails.price" },
    minimumPrice: { $min: "$movieDetails.price" },
    maximumPrice: { $max: "$movieDetails.price" }}}
])
< {
  _id: 'Avatar',
  totalTicketsSold: 3,
  totalRevenue: 1050,
  averagePrice: 350,
  minimumPrice: 350,
  maximumPrice: 350
}
{
  _id: 'Inception',
  totalTicketsSold: 3,
  totalRevenue: 900,
  averagePrice: 300,
  minimumPrice: 300,
  maximumPrice: 300
}
```

>_MONGOSH

```
> db.bookings.aggregate([
  {$lookup: {
    from: "movies",
    localField: "movie_id",
    foreignField: "_id",
    as: "movieDetails"}}},
  { $unwind: "$movieDetails" },
  {$lookup: {from: "customers",
    localField: "customer_id",
    foreignField: "_id",
    as: "customerDetails"}}},
  { $unwind: "$customerDetails" },
  {$group: {
    _id: "$customerDetails.name",
    totalMoviesBooked: { $sum: 1 },
    totalSpent: { $sum: "$movieDetails.price" },
    averageSpending: { $avg: "$movieDetails.price" },
    moviesWatched: { $push: "$movieDetails.title" }}}
])
< {
  _id: 'Priya',
  totalMoviesBooked: 1,
  totalSpent: 300,
  averageSpending: 300,
  moviesWatched: [
    'Inception'
  ]
}
```

>_MONGOSH

```
> db.bookings.aggregate([
  {$group: {
    _id: "$customer_id",
    totalBookings: { $sum: 1 }},
  {$match: {totalBookings: { $gt: 1 }}},{
    $lookup: {
      from: "customers",
      localField: "_id",
      foreignField: "_id",
      as: "customerDetails"
    }
  },

  { $unwind: "$customerDetails" },

  {
    $project: {
      _id: 0,
      customerName: "$customerDetails.name",
      totalBookings: 1
    }
  }
])
< {
  totalBookings: 2,
  customerName: 'Kiran'
}
{
  totalBookings: 2,
```

```
> db.bookings.aggregate([
  {$lookup: {
    from: "movies",
    localField: "movie_id",
    foreignField: "_id",
    as: "movieDetails"
  }},
{ $unwind: "$movieDetails" },
{$group: {
  _id: "$movieDetails.genre",
  totalBookings: { $sum: 1 }}}
])
< {
  _id: 'Fantasy',
  totalBookings: 3
}
{
  _id: 'Sci-Fi',
  totalBookings: 5
}
```

>_MONGOSH

```
> db.bookings.aggregate([
  {$lookup: {
    from: "movies",
    localField: "movie_id",
    foreignField: "_id",
    as: "movieDetails"}}},
{ $unwind: "$movieDetails" },
{$lookup: {
  from: "customers",
  localField: "customer_id",
  foreignField: "_id",
  as: "customerDetails"
}},
{ $unwind: "$customerDetails" },
{$group: {
  _id: "$movieDetails.title",
  totalBookings: { $sum: 1 },
  customerList: { $push: "$customerDetails.name" }}}
])
< {
  _id: 'Interstellar',
  totalBookings: 2,
  customerList: [
    'Rahul',
    'Kiran'
  ]
}
{
  _id: 'Inception',
```